



Application Security Log Output Standard

STANDARD NUMBER: DAS-ITS-2100-12-B	EFFECTIVE DATE: 12/22/2017	APPROVED BY: Anupam Srivastava, State Chief Information Security Officer
REPLACES STANDARD DATED: n/a	AUTHORITY: Ohio Revised Code 125.18 Office of Information Technology - duties of director - contracts, Ohio IT Standard ITS-SEC-02, "Security Controls Framework"	

1.0 Purpose

The purpose of this standard is to provide requirements to Ohio Department of Administrative Services (DAS) technical teams that will allow for the consistent implementation of application audit log output standards.

2.0 Scope

The scope of this standard includes all DAS applications. This standard represents the minimum acceptable criteria governing audit log output standards.

3.0 Standard

3.1 Log entries should capture user access events and, as a minimum, include what is listed in the table below as required.

Field	Description	Entry
Application	Name of the application generating the log	Required
Date	The date an event occurred (format should be standardized, such as DD-MM-YYYY or MM-DD-YYYY)	Required
Time	The time the event occurred	Required
Time Zone	GMT time and offset	Required
User Name	The name of the user accessing the application or attempting to access the application	Required
Person	The name/identifier of the person whose CPI was accessed	Required
Version	The version of the application	Recommended
Event	The event type (e.g., alarm, warn, information)	Recommended

Field	Description	Entry
Access	The access type (e.g., read, write, update)	Recommended
Command	The command, options and parameters, directly initiated by the user	Optional
Resource	The resource (e.g., database, file) accessed	Optional
Record	A “before” and “after” picture of each modified record (or the data element(s) changed within a record)	Optional

3.2 Exception Process

To request an exception to one or more of the requirements outlined in this standard, please complete an IT Security Exception Request Form.

3.2.1 The form is located within the [IT Enterprise Services Portal](#) under the "Services & Products" category.

3.2.2 If you have questions, please contact the DAS Office of Information Security and Privacy (refer to Section 9.0 Inquires for contact information).

4.0 References

- 4.1 **Ohio IT Standard ITS-SEC-02, Security Controls Framework:** This state IT standard specifies the minimum requirements for information security in all agencies and identifies the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53 as the framework for information security controls implementation for the state.
- 4.2 **National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53, Security and Privacy Controls for Federal Information Systems and Organizations:** NIST SP 800-53 provides guidelines for selecting and specifying security controls for federal government information systems.
- 4.3 **DAS Policy 2100-12, Audit and Accountability Policy:** DAS Policy 2100-12 defines information system audit and accountability requirements that will assist in assessing the adequacy of system controls, ensuring compliance with established policies and operational procedures, and uniquely tracing the actions of system users.
- 4.4 **DAS IT Standard DAS-ITS-2100-01-A, Password Standard for Organizational Users:** Provides password requirements to DAS technical teams that will allow for the consistent implementation of password standards across DAS-managed system assets for internal users (e.g., state employees, contractors, county workers, etc.).
- 4.5 **IRS Publication 1075, Tax Information Security Guidelines for Federal, State and Local Agencies:** Provides guidance to ensure the policies, practices,

controls, and safeguards employed by recipient agencies, agents, or contractors adequately protect the confidentiality of federal tax information.

5.0 Definitions

<i>Audit</i>	A review and examination of information system records and activities to assess the adequacy of system controls, to ensure compliance with established policies and operational procedures, and to recommend necessary changes in controls, policies, or procedures. ¹
<i>Audit Logs</i>	A record showing who has accessed a computer system and what operations have been performed during a given time period. Audit logs are useful both for maintaining information technology security and recovering lost transactions.
<i>Confidential Personal Information (CPI)</i>	Personal information that falls within the scope of section 1347.15 of the Ohio Revised Code and that an agency is prohibited from releasing under Ohio's public records law.
<i>User</i>	An individual or (system) process authorized to access an information system. ²

6.0 Related Resources

Document Name
<i>DAS IT Operations and Management Policy 2100-01, "Password Policy"</i> https://das.ohio.gov/technology-and-strategy/policies
<i>DAS IT Standard ITS-2100-01-A, "Password Standard for Organizational Users"</i> https://das.ohio.gov/technology-and-strategy/policies
<i>Ohio IT Standard ITS-SEC-01, "Data Encryption and Cryptography"</i> https://das.ohio.gov/technology-and-strategy/policies
<i>Ohio IT Standard ITS-SEC-02, "Security Controls Framework"</i> https://das.ohio.gov/technology-and-strategy/policies
<i>NIST SP 800-53; Security and Privacy Controls for Federal Information Systems and Organizations</i> https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-53r4.pdf

¹"NIST Special Publication 800-32, Introduction to Public Key Technology and the Federal PKI Infrastructure," U.S.Department of Commerce National Institute of Standards and Technology, February, 2001. <<https://csrc.nist.gov/publications/nistpubs/800-32/sp800-32.pdf>>.

²"FIPS 200, Minimum Security Requirements for Federal Information and Information Systems." Federal Information Processing Standards Publication (FIPS) March 2006. <<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.200.pdf>>.

7.0 Implementation

7.1 DAS IT system managers are expected to work towards implementation of the application security log output requirements outlined in this standard as quickly as possible.

7.1.1 For new DAS-managed systems, the application security log output requirements outlined in this standard shall immediately become requirements.

7.1.2 For existing DAS-managed systems and DAS-managed systems that have moved beyond the requirements phase, DAS IT system managers have one year from the effective date of this policy to implement the requirements.

7.1.3 If for some reason a DAS IT system manager feels that it is not technically feasible to implement the application security log output requirements outlined in this standard, an exception request shall be submitted in accordance with Section 3.2.

8.0 Revision History

Date	Description of Changes
12/22/2017	Original Standard
11/28/2018	Updated the exception request language to align with the current procedure.
02/24/2020	Updated standard template.
06/22/2021	Conducted routine maintenance review of citations and references.
06/22/2022	Scheduled standard review.

9.0 Inquiries

For information regarding this DAS IT standard, please contact:

Office of Information Security & Privacy
Office of Information Technology
Ohio Department of Administrative Services
30 East Broad Street, 19th Floor
Columbus, Ohio 43215

Telephone: 614.644.9391

Email: state.isp@das.ohio.gov

Web: <https://das.ohio.gov/technology-and-strategy/information-security-privacy>

DAS IT Standards are online at:

<https://das.ohio.gov/technology-and-strategy/policies>

10.0 Attachments

None.